

Project (Car Price Prediction)

Dataset:

Attribute Name	Definition	Example
ID	unique vehicle identifier	45813960, 45730073
Levy	Taxes and fees	891, 730
Manufacturer	Manufacturer Company	MERCEDES-BENZ, OPEL
Model	The model of the vehicle	CLA 250, Astra
Prod. year	Year of production	2021, 2008
Category	Vehicle category	Sedan, Hatchback
Leather interior	Is the interior designed with leather	Yes or No
Fuel type	Type of fuel the car work with	Fuel, Hybrid
Engine volume	Engine volume	3.5, 2.0 Turbo
Mileage	How many KMs the car has walked	10573 km, 204000 km
Cylinders	number of cylinders	4, 8
Gear box type	Transmission type	Automatic, Manual, Tiptronic
Drive wheels	Drive wheels	Front, Rear
Doors	Number of doors	
Wheel	steering wheel position	Left Wheel, Right-hand drive
Color	Car Color	White, Green
Airbags	Number of Airbags	4, 6
Price (Target column)	Car Price	41281, 6272

Project Requirements:

1. Preprocessing: Before building your classification models, you need to make sure that the dataset is processed, clean and ready-to-use.
2. Regression:
 - Train at least 2 different regression models.
 - Choose at least two hyperparameters to vary, and explain the impact of varying those two hyperparameters with different values.
 - Train the models on the given training set (train_data.csv) and evaluate the model on the given testing set (test_data.csv).
3. Report:
 - You must explain in detail the preprocessing techniques that you applied on the dataset before the modeling step.
 - Perform analysis and visualizations on the dataset and explain how features affect each other.
 - State the models and hyperparameters used in the modeling step
 - Mention any other techniques that you used to enhance the results.
 - Write a final conclusion.

Note:

- Any additional models and not trivial preprocessing techniques that are not discussed in labs will be considered a bonus.
- Good GUI/deployment will also be considered a bonus.
- A portion of the marks will be based on individual evaluation.